

DELTA, TEB4810 – 3U 48V/100Ah

NOMINAL

Voltage	48.0V _{DC}
Nominal Capacity (25°C)	100Ah (capacity is measured under to 0.2C / 25°C @ BOL)

ELECTRICAL

Energy	
- Nominal Energy (25°C, 0.2C)	4800Wh (capacity is measured under to 0.2C / 25°C @ BOL)
- Volumetric Energy Density	184Wh/L
- Gravimetric Energy Density	123Wh/kg
Standard Discharge (25°C)	
- Maximum continuous Current	100A
- Cut-off Voltage	37.5V _{DC}
- LVBD Voltage	41.3V _{DC}
Standard Charge (25°C)	
- Maximum continuous Current	100A
- Recommended Charging Current	50A
- Charge Voltage	Max 53.5 VDC (Recommend: 51.9VDC)
Charge current limit function	10A
Internal impedance	<19mΩ

MECHANICAL

Dimensions (W x H x D)	Without handle: 442 (±1) x 131 (±1) x 450 (±1) mm With handle & terminal: 482 (±1) x 131 (±1) x 491 (±1) mm
Weight	approx. 39.0±2.0kg
Total Cells Quantity in Battery Module	15pcs of cell in series (Prismatic type cell)
Materials of Battery case	Carbon steel with corrosion resistant coating
Cell	LFP 100Ah
Max Quantity of Battery Parallel Connection	15 for one RS-485 and max 30 pcs in system.
Color	RAL9003 (White)

ENVIRONMENTAL

Calendar Life (25°C)	> 10 years
Cycle Life (1C/1C, DOD 100%, 25°C)*	SOH 80%, 5000 cycles
Operating Temperature**	0 to +50 °C (De-rating from 35 °C)
Storage Temperature	0 to +45 °C (Recommended range: 10 to +30 °C)

BMS PARAMETERS

Voltage (Charge)	Cell voltage protection: 3.75V protection (Recover at 3.45V) Total voltage protection: 55.5V protection (Recover at 50.4V)
Voltage (Discharge)	Cell voltage protection: 2.5V protection (Recover at 3.1V) Total voltage protection: 37.5V protection (Recover at 45V)
Current (Charge)	Normal ≤ 100A Charging current limiting function: > 110A (10sec) Charging current limiting function: > 125A (3sec)
Current (Discharge)	Normal ≤ 100A Over current protection 1: > 110A and < 125A (10sec) Over current protection 2: > 125A (3sec) Short circuit protection: ≥ 320A (<1ms)
Temperature (Cell)	Low temperature protection: Charging < 0°C, Discharging < -20°C High temperature protection: Charging > 55°C, Discharging > 60°C
Temperature (PCB)	Temperature protection ≥95°C (Recovery < 75°C)
Cell Balance (Balance)	BMS include Passive balancing circuit and algorithm , (Current: < 100mA)
Dry contact	D.O x 2